



Demonstrate:
a new app to help
you create
iOS apps

> code / creativity / community >

devworx >

Creating stunning visualisations using R

>>Data visualisation is coming up fast and strong. Try your hand at it if you don't want to be left behind>

What is data visualisation?

Put simply, data visualisation is the pictorial representation of data. Any data set can be understood at a glance by representing it on interactive charts and graphs. This presentation method is employed everywhere around you, right from statistical graphs in textbooks and academic papers to colourful infographics in newspapers. As technology has progressed, we've been able to add increasingly more specifics and data, and with more data arises the need for accurate and appropriate visualisation techniques. That's where 'R' comes in handy.

What is R?

R is a language developed primarily for statistical computing and data analysis. Data analysis and data graphics go hand in hand so it's imperative for a language of R's nature to have support, which will help it employ some amazing graphic techniques for data visualisation.

Programming to create visualisation?

Is R user friendly for students?

R, since its introduction, has become quite popular not only among the PhD holders in Data Science but also amongst mainstream users in need of a common platform for computing and visualisation. In this workshop, we'll get you started on some simple data visualisation techniques using R, which will serve as an adequate platform to develop visualisations that are even more advanced. For first time users of programming languages: Do not be afraid. The idea of coding from scratch can be intimidating, especially if you've never used programming languages before. Remember that every line of code is simply a written equivalent of a keyboard shortcut in a sense. It performs a task and these commands can mostly be looked up in a book or online.

What are the advantages of using R over MATLAB?

R has some distinct advantages. Since it's a free software service released under the GNU license it's free! This gives it a huge advantage over commercial software such as MATLAB, which are really expensive especially for students and small enterprises. Because of its Open Source nature R attracts a host of talented developers

Year	Runs	Average	100s	50s
2009	215	35.82	0	2
2008	379	41.44	1	2
2007	78	19.5	0	0
2006	419	45.9	3	0
2005	840	91.82	2	5
2004	730	70	2	8
2003	58	20	0	1
2002	523	41.52	2	2
2001	1000	62.5	4	3
2000	647	80.87	3	1
1999	1088	68	6	4
1998	575	63.88	2	1
1997	1009	62.68	3	6
1996	1392	55.68	4	5
1995	139	17	0	1
1994	915	91.5	3	2
1993	844	64.6	1	8
1992	267	24.27	0	1
1991	776	55.42	2	6
1990	1085	48.31	4	5
1989	541	87.82	2	3
1988	1562	78.1	7	5
1987	756	47.25	1	5
1986	897	28.8	0	2
1985	276	34.5	0	2

A screenshot from the data sheet where Sachin Tendulkar's records are stored. Source: stats.espncricinfo.com

which means awesome support, documentation and user defined packages for diverse needs. R has some amazing packages to unlock a variety of complex and useful data visualisation techniques.

What else can R do?

Besides creating stunning visualisations you can also easily perform run-of-the mill data analysis. There's good support for computations regarding statistics and probability, and support for machine learning and data mining is getting better.

Installing the IDE

One of the best and most robust IDEs for R is

RStudio. (An IDE or integrated development environment is a facilitator that helps you communicate effectively and intuitively with the programming language). It comes bundled with R software and basic packages, and can be installed on Linux, Windows or Mac with the single click of a button.

The installation in Windows/Mac is a routine procedure as with any other software. Download RStudio Desktop, run the .exe file and follow the procedural steps. While in Linux one has to install R via the terminal (type 'sudo apt-get install r-base') and then install RStudio separately by selecting it from the software centre.

Basic layout – as user-friendly as it gets!

The home screen is an arrangement of four panes, each of which can be expanded to fill the screen whenever required. The top left